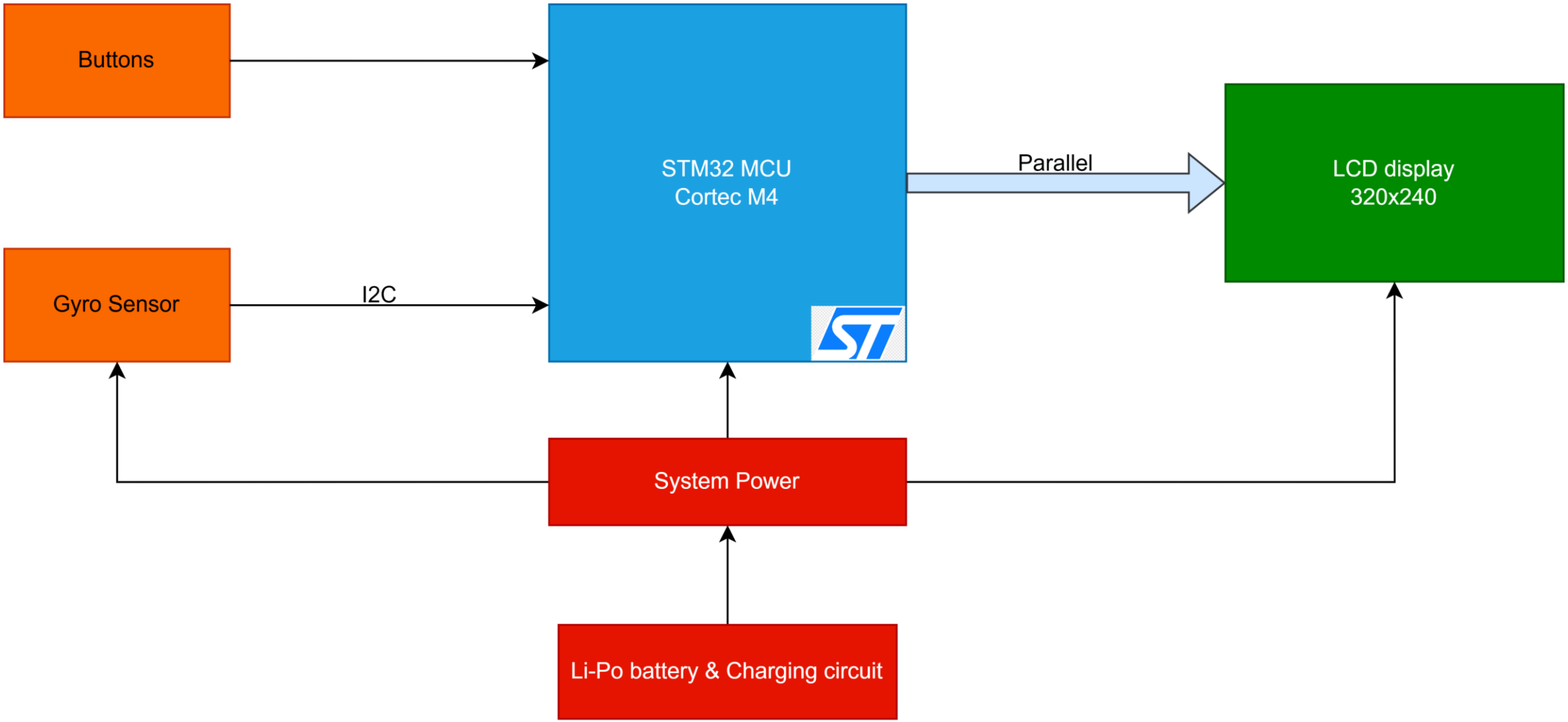


GamePal

(A hand-held gaming companion)

-by Atharv More & Mayuresh Pitale

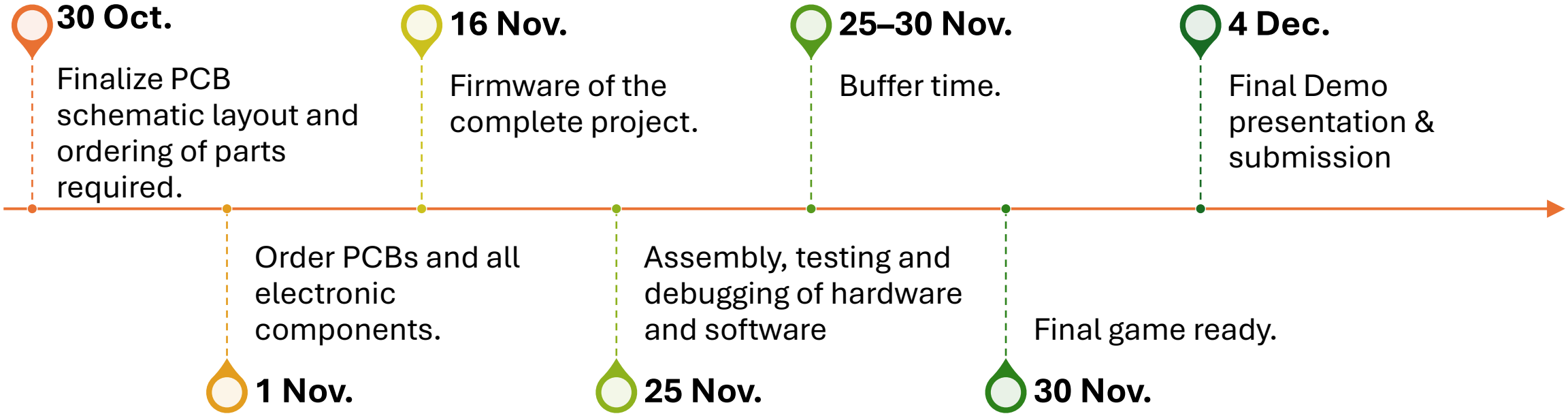
Hardware Elements



Software Elements

- **Software Architecture Overview**
- **Main Game Loop:**
  - Initializes hardware and game state.
  - Continuously reads inputs, updates game logic, and renders the display.
- **New Software Modules:**
  - **LCD Driver:** Manages screen initialization and pixel buffer updates.
  - **Gyro Driver:** Reads and interprets angular velocity data for steering input.
  - **GPIO Handler:** Manages button press detection with debouncing.
  - **Game Logic:**
    - Player Control: Translates gyro data and button presses into motorcycle movement.
    - Obstacle Engine: Procedurally generates and manages obstacles on the track.
    - Physics & Collision: Handles collision detection between the player and obstacles.
    - UI Manager: Renders the score, speed, and game-over screens.

Key Milestones & Deliverables



Deliverables & Fall-back Plan

- **Project Deliverables**
- A fully functional, custom-built handheld moto racing game.
- Complete PCB design files for the custom PCB.
- Well-documented source code for the STM32 microcontroller.
- A final demonstration video of the working prototype.
- Future Scope: External memory for additional games.
- **Fall-back Plan**
- In the event of critical PCB manufacturing errors or hardware failures, the project will be implemented on a STM32F411E-disco board. All peripherals (LCD, gyro, buttons) will be connected via a zero PCB to ensure the software and gameplay functionality can still be delivered.